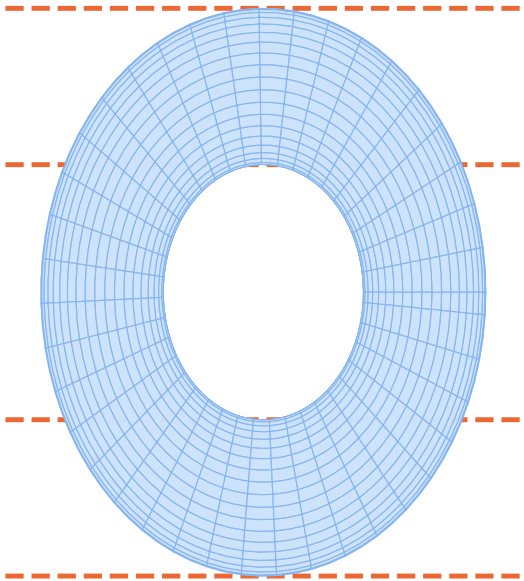
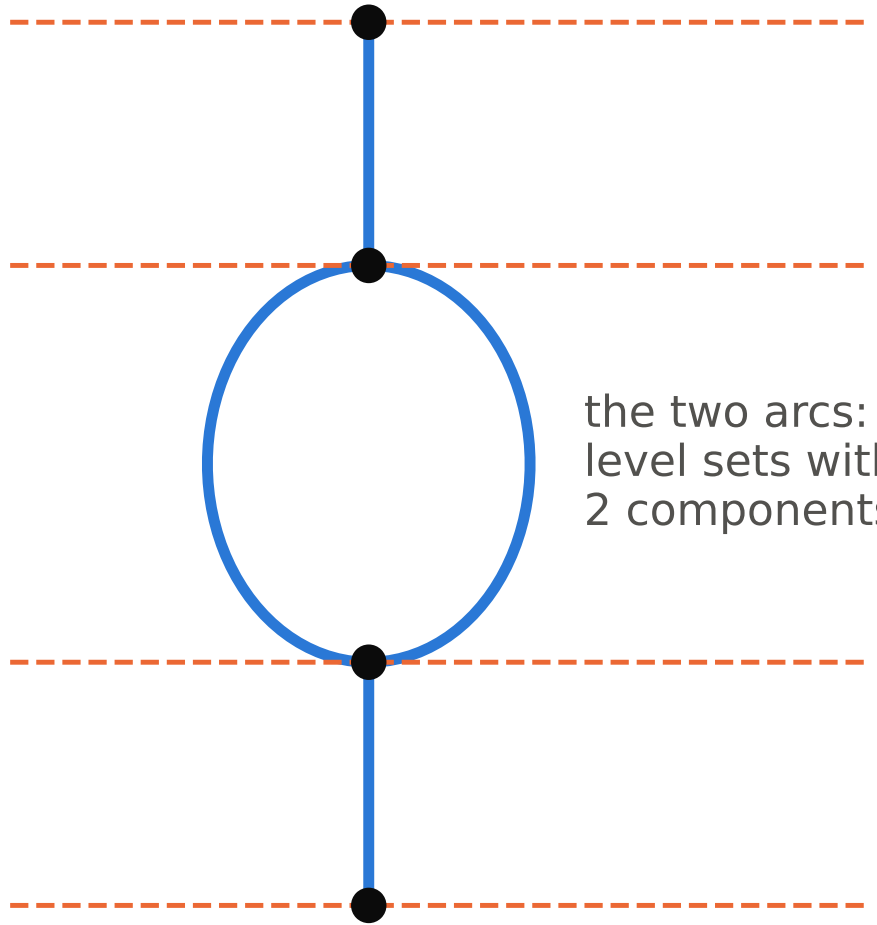
$X$ = torus, $f$ = height



Reeb graph $R_f(X)$



the two arcs:
level sets with
2 components